One Encounter, Two Readers: Verified Clinical Notes for Doctors and Faithful Summaries for Patients

Jyothika Goparaju
2025701007

Namrata Mokshagundam
2025701011

Repaka Doshik Rahul
2026701035

Surya Teja Thodupunuri
2025701054

1 Introduction

Clinical documentation systems increasingly use Large Language Models (LLMs) to generate structured medical notes from patient-doctor conversations. A clinical note serves two readers: doctors, who rely on it for clinical decision-making, and patients, who need an understandable account of their consultation. Therefore, the generated documentation must be both **factually reliable and sufficiently complete**. However, existing approaches face **two important limitations**. First, **post-hoc verification methods** identify missing or unsupported information only after the consultation is complete. Since the patient is no longer available to clarify missing details, the system can only remove unsupported statements or risk introducing information that was never provided. Second, **conventional entailment-based verification** treats every statement in a clinical note as a factual claim. This does not account for the distinction between directly reported facts in the Subjective and Objective (S/O) sections and clinical inferences in the Assessment and Plan (A/P) sections. Consequently, valid clinical reasoning may be incorrectly rejected because it is not explicitly stated in the original conversation. To address these limitations, this project proposes a **verification-driven clinical documentation framework** that combines interactive information recovery through **targeted questioning** with **section-aware verification**. The framework aims to produce evidence-linked SOAP notes for doctors while generating faithful, accessible summaries for patients from the verified record.

2 Related Work

Building on these challenges, existing research has explored both post-hoc verification of clinical summaries and interactive approaches to clinical information collection. Nagar et al. (2025) proposed **uMedSum**, a unified framework for **clinical abstractive summarization** that addresses confabulations and information omissions through a three-stage pipeline. It generates an initial summary, applies atomic-claim decomposition and Natural Language Inference (NLI) to identify unsupported or contradictory statements, and finally detects and incorporates relevant information missing from the summary through coverage-based selection and merging. While this approach improves the factual consistency and completeness of generated summaries, verification takes place after the source dialogue is complete. Consequently, it cannot recover missing information by asking the patient targeted questions during the consultation.

An alternative direction focuses on **improving information collection** through **interactive clinical intake**. Chen et al. (2026) introduced **Aegle**, a collaborative multi-agent framework in which an **orchestrator** coordinates **specialist agents** to gather patient information and construct a structured clinical representation. Specialists contribute questions, while an aggregator consolidates their responses to guide the intake process. Although this enables structured information collection through interaction, it does not independently verify whether the statements in the resulting clinical note are supported by the collected evidence. Thus, while uMedSum focuses on post-generation verification and Aegle on interactive intake, neither addresses the integration of verification-guided information recovery with section-specific assessment of clinical statements.

3 Proposed Method

The proposed framework integrates **multi-agent clinical intake** with **verification-guided information recovery** and **section-aware note verification**. Figure 1 shows the intended system flow. Patient responses are consolidated into an evidence-linked SOAP state. Before intake is frozen, **Verify-and-**

Ask (RQ1) identifies missing or unsupported information in the Subjective and Objective (S/O) sections and uses targeted questions to recover missing patient-reported facts. After the freeze, the Assessment and Plan (A/P) sections undergo **section-aware verification (RQ2)**, which evaluates clinical statements using their supporting evidence chains. The resulting **doctor-facing note** contains evidence links and flags for review and sign-off. Following sign-off, a **patient-facing summary** is generated from the verified note.

3.1 Verify-and-Ask (RQ1)

Verify-and-Ask addresses information gaps that conventional post-hoc verification cannot recover once a consultation has ended. Before the intake is frozen, the generated Subjective and Objective (S/O) sections are decomposed into atomic claims and checked against section-specific evidence. Subjective claims are verified against relevant patient dialogue turns, while Objective claims are checked against the encounter's measurement and examination records. Each claim is classified as supported, contradicted, or unsupported, allowing the system to distinguish incorrect information from missing information.

Based on these verdicts, fixed controller rules determine the next action. Supported claims are retained, while contradicted claims are corrected using the cited evidence and re-verified. Missing or unsupported mandatory patient-reported information, such as symptoms, duration, medication, or allergies, triggers a targeted clarification question. Objective gaps are flagged for the doctor, as these cannot be established through patient responses alone. Other unrecoverable claims are removed or marked as uncertain. The patient's response is subsequently checked against the claim before being added to the note, with a maximum budget of three clarification questions in the proposed design. This allows verification to guide information collection rather than merely filtering the completed note. The current implementation includes question generation and filtering.

3.2 Section-Aware Verification (RQ2)

Unlike S/O statements, which describe reported symptoms and observed findings, the Assessment and Plan (A/P) sections contain clinical inferences, including possible diagnoses and treatment decisions. Such statements may not be directly entailed by individual dialogue turns, making uniform NLI-based verification unsuitable. To address this, the proposed framework evaluates each A/P statement through its supporting evidence chain. The statement is linked to the clinical features on which it is based, along with relevant retrieved reference passages when available. A verifier then determines whether the evidence sufficiently supports the statement, classifying the chain as valid or broken. Statements with broken evidence chains are flagged for doctor review rather than automatically deleted, since rejecting a clinical assessment requires human judgment. Following verification and doctor sign-off, a patient-facing summary is generated from the signed note, excluding unresolved flagged information and checking each summary statement against the approved record. The RQ2 verifier and patient-facing summary generation are planned extensions.

4 Implementation

4.1 Dataset Preparation

Experiments use the Ambient Clinical Intelligence Benchmark (ACI-Bench) (**?**), which contains 207 doctor–patient encounters with corresponding clinical notes. The corpus includes 67 training encounters, 20 validation encounters, and three test subsets containing 40 encounters each. For the uMedSum baseline, three training encounters were reserved as in-context learning examples, while 140 encounters from the validation and test subsets were used for evaluation.

A separate set of simulated consultations was constructed for the proposed pipeline using 110 encounters from ACI-Bench. These comprised 67 training encounters, 20 validation encounters, and 23 encounters from the first test subset. Rather than directly using the original dialogues as inputs to the proposed pipeline, each encounter was processed through an Aegle-inspired multi-agent intake system implemented using LangGraph and Qwen3-14B.

For each encounter, a Standardized Patient Agent was initialized using the corresponding original ACI-Bench dialogue as its source of information. It simulated patient responses to questions raised during the consultation, while being constrained to information available in the original dialogue. The agent was not permitted to invent clinical details such as ages, medication doses, dates, examination findings, allergies, or negative findings that were absent from the source. Unavailable

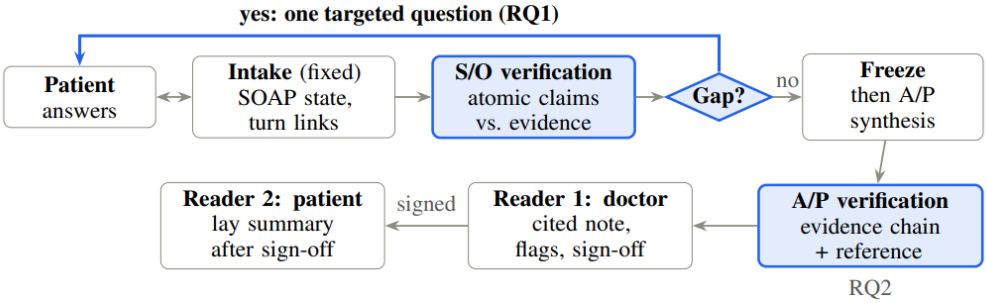


Figure 1: Overview of the proposed verification-driven clinical documentation framework. RQ1 performs verification-guided information recovery before intake freeze, while RQ2 evaluates clinical inferences after the freeze.

information was treated as unknown rather than assumed to be absent.

The simulated consultation was coordinated by an Orchestrator, which determined the information-collection requirements and activated relevant Specialist Agents. The specialists operated in parallel to identify relevant clinical information, propose structured updates, and formulate targeted follow-up questions. Their outputs were consolidated by an Aggregator, which maintained the structured patient record and determined the next question or action. The record was organized into five case-feature groups: basic information, present illness, past history, physical examination, and auxiliary examination findings. The consultation proceeded for up to 15 turns, after which the Aggregator produced a structured SOAP note containing the collected clinical information, preliminary diagnosis, and treatment plan.

The 110 simulated encounters were divided into development and test sets using a fixed random seed of 42, with 20% assigned to development. This resulted in 22 development cases and 88 test cases. The development set was used for selecting verification and omission-recovery thresholds, while the test set was reserved for final evaluation. The simulated consultation transcripts served as inputs to the proposed pipeline. Reference S/O information was maintained separately and used only for evaluation, without being supplied to note generation, verification, or omission-recovery components.

4.2 Models and Pipeline

The uMedSum framework (**?**) was reproduced and adapted for full SOAP note generation as the post-hoc verification baseline. The pipeline uses LLaMA-3-8B for element-aware note generation with one-shot in-context learning, followed by DeBERTa-v3-large for NLI-based confabulation detection and removal using Recursive Threshold-Based Text Segmentation (RTB-TS). The final stage uses all-MiniLM-L6-v2 and KeyBERT to identify missing information, with GPT-2 for perplexity-guided sentence insertion.

For interactive intake, an Aegle-inspired multi-agent workflow was implemented using LangGraph and Qwen3-14B. An orchestrator coordinates specialist agents that propose questions in parallel, while an aggregator consolidates their responses into a structured patient record. A standardized patient agent, grounded in the corresponding ACI-Bench dialogue, responds to these questions for up to 15 turns. The resulting consultation dialogue and SOAP note are then used as inputs to the proposed verification pipeline.

The proposed Verify-and-Ask pipeline uses LLaMA-3-8B for initial SOAP generation and DeBERTa-v3-large for claim verification. Sentence-first verification checks a complete sentence before decomposing it into atomic claims when the sentence is not sufficiently supported. Stage 3a uses MiniLM-based coverage scoring to identify possible omissions, an LLM to rewrite candidate information into third-person clinical statements, and NLI and similarity checks to validate additions and reject duplicates. Stage 3b generates targeted questions for unresolved gaps and filters questions that are already answered or redundant. In the current implementation, the A/P section is passed through unchanged; this protects clinical inferences from inappropriate extractive filtering but does not constitute RQ2 verification.

4.3 Prompting Strategy

Table 1 summarizes the main prompts used across the three pipelines. The prompts enforce section-aware extraction, evidence-grounded generation, and conservative handling of unavailable information.

4.4 Verification and Recovery Rules

The main thresholds used by the implemented S/O pipeline are summarized in Table 2. For Stage 2, a claim with entailment probability at least 0.90 is retained. A contradiction probability of at least 0.50, together with entailment below 0.90, triggers correction using dialogue evidence. Claims with intermediate entailment scores are retained as uncertain and flagged, while claims below 0.50 are removed from the active note. For Stage 3a, candidate omissions are identified using a maximum cosine similarity of 0.40 against the verified note.

5 Experimental Setup

5.1 Evaluation Splits

The baseline and proposed pipeline use separate evaluation setups and are therefore not directly comparable as a single benchmark. The baseline evaluates 140 original ACI-Bench dialogues and full SOAP references, with three training examples reserved for one-shot prompting. The proposed pipeline evaluates 110 simulated consultation outputs, partitioned into 22 development and 88 test cases, and focuses on the S/O sections. The development split is used for threshold selection; the test split is used for the primary reported results. Reference notes are not passed to the generation or verification components.

5.2 Evaluation Metrics

We report lexical, semantic, faithfulness, and task-specific clinical metrics. ROUGE-1 and ROUGE-2 measure unigram and bigram overlap with the reference, while ROUGE-L and ROUGE-Lsum measure longest-common-subsequence overlap at the sequence and summary levels. BERTScore computes semantic similarity using contextual token embeddings. SummaC-Conv estimates factual consistency without a reference summary by applying sentence-level NLI. QuestEval estimates informativeness and faithfulness through question generation and answering between the source and generated note.

For the proposed pipeline, NLI entailment estimates whether a generated claim is supported by the dialogue. Atomic fact recall measures the proportion of reference facts represented in the generated note; critical fact recall restricts this calculation to clinically important facts. Recoverable fact recall considers reference facts that are supported by the dialogue. Fact recovery rate measures the fraction of genuine omissions after Stage 2 that are recovered by Stage 3a. Supported rate, unsupported claim rate, and contradiction rate characterize grounding quality. Gap-detection precision and recall measure omission candidate identification, while duplicate-addition and question-redundancy rates quantify unnecessary insertions and questions.

6 Results and Analysis

6.1 uMedSum Baseline

Table 3 presents the reproduced uMedSum results on 140 full-SOAP encounters. These are results from our adapted reproduction and not the original paper's reported numbers. Stage 2 increases mean NLI entailment from 0.7969 to 0.9080, indicating improved grounding after confabulation removal. However, average note length decreases from 353.2 to 171.4 words, accompanied by a decline in ROUGE-1 F1 from 0.4671 to 0.3365. This illustrates the trade-off between conservative claim filtering and information coverage.

Stage 3 partially restores coverage: average length increases to 201.4 words, ROUGE-1 increases to 0.3572, and QuestEval increases from 0.3503 to 0.3742. However, the final scores remain below the initial generation for several overlap metrics. The reproduced pipeline also evaluates full SOAP notes, whereas the original uMedSum results shown for comparison evaluate A/P sections. Therefore, the values should be treated as contextual reference points rather than a directly comparable benchmark.

6.2 Multi-Agent Intake

The Aegle-inspired intake system generated 110 simulated consultations using Qwen3-14B and LangGraph. The orchestrator selected specialist agents, specialists proposed questions and structured updates in parallel, and the aggregator consolidated these into the patient record. The standardized patient agent was restricted to information available in the original ACI-Bench dialogue and

Pipeline	Prompt / Agent	Main instruction
uMedSum	Element extraction	Extract Subjective, Objective, Assessment, and Plan elements from the dialogue, including symptoms, history, examination, investigations, diagnoses, and treatment.
uMedSum	SOAP synthesis	Generate a complete SOAP note from extracted elements, following a one-shot demonstration and preserving the required section structure.
Multi-agent intake	Orchestrator	Select history-taking or diagnosis-and-plan mode and route the case to relevant specialist agents.
Multi-agent intake	Specialist agents	Identify relevant clinical information, propose structured case-feature updates, and suggest a targeted next question.
Multi-agent intake	Aggregator	Consolidate specialist outputs, update the structured patient record, and formulate the next question.
Multi-agent intake	Standardized patient	Answer only from the original ACI-Bench dialogue; do not invent unavailable ages, doses, dates, findings, or negative responses.
Verify-and-Ask	SOAP generation	Use only explicit consultation information; do not invent symptoms, measurements, examination findings, medications, diagnoses, or plans.
Verify-and-Ask	Atomic decomposition	Split compound statements into atomic claims while preserving negation and the original meaning.
Verify-and-Ask	Contradiction correction	Rewrite a conflicting claim using only supporting dialogue evidence; return `NONE` if no grounded correction is possible.
Verify-and-Ask	Omission rewriting	Convert patient statements into concise third-person clinical statements and retry if the rewrite introduces unsupported details.
Verify-and-Ask	GapQuestioner	Generate targeted questions for unresolved gaps and remove questions that are already answered or duplicate earlier questions.

Table 1: Summary of the principal prompting components.

could not supply unavailable clinical details.

An implementation issue was observed when information such as exact medication names or dosages was absent from the source dialogue. The intake workflow could repeatedly ask for the same unavailable detail across multiple turns. This observation motivated the question-answer and redundancy filters in Stage 3b. It is an implementation finding and should not be interpreted as a measured improvement in clinical outcomes.

6.3 Verify-and-Ask

The proposed pipeline is evaluated on 88 test cases. We compare three outputs: (A) initial S/O note from Stage 1, (B) verified S/O note after Stage 2, and (C) final S/O note after Stage 3a omission recovery. Table 4 reports the principal results.

Stage 2 improves grounding: supported rate increases from 0.9590 to 0.9862, unsupported claim rate decreases from 0.0334 to 0.0031, and contradiction rate decreases from 0.0273 to 0.0000. The valid-claim retention proxy is 98.99%, suggesting that the sentence-first strategy largely preserves dialogue-supported facts while removing unsupported claims.

Stage 3a improves S/O coverage. Atomic fact recall increases from 0.6334 to 0.7342, critical fact recall from 0.6426 to 0.7235, and recoverable fact recall from 0.7101 to 0.8199. S/O ROUGE-1 F1 increases by 5.11 points and ROUGE-Lsum by 3.97 points between A and C. The reported unsupported-addition rate is 0.0% on the test split after rewriting and NLI validation. These results indicate that validated omission recovery improves coverage in the available dialogue without adding statements classified as unsupported by the evaluation procedure.

Table 5 reports task-specific diagnostics. Stage 3a identifies 32.47% of genuine omissions and recovers 42.82% of the omissions remaining after Stage 2. The gap-detection precision is 13.89%, indicating that the initial coverage-based candidate detector is permissive and relies on subsequent rewriting and validation to reject unsuitable candidates. The duplicate-addition rate is 7.41%. Stage 3b retains 88 questions across 88 test encounters, while 36.23% of generated questions are filtered as already answered or redundant.

Importantly, Stage 3b currently generates and filters questions; it does not yet conduct live patient interaction or incorporate patient answers. Thus, the reported fact recovery results correspond to Stage 3a recovery of information already present in the dialogue, not to successful recovery through a new patient response. Similarly, A/P is unchanged in all cases because RQ2 verification has not yet been implemented. The 100% unchanged rate is an isolation safeguard, not evidence of successful A/P verification.

An ablation on the 22-case development split compared atomic-only verification with sentence-first verification followed by atomic decomposition when required. Atomic-only verification scored

Component	Threshold / rule	Action
Stage 2	Entailment ≥ 0.90	Retain claim.
Stage 2	Contradiction ≥ 0.50 and entailment < 0.90	Attempt evidence-based correction and re-verification.
Stage 2	Entailment 0.50–0.90	Retain as uncertain and flag.
Stage 2	Entailment < 0.50	Remove unsupported claim.
Stage 3a	Coverage cosine ≤ 0.40	Flag candidate omission.
Stage 3a	Entailment ≥ 0.80, contradiction ≤ 0.30	Accept grounded rewrite.
Stage 3a	Entailment ≥ 0.70 or cosine ≥ 0.85	Reject duplicate addition.
Stage 3b	Maximum questions $= 5$	Cap generated questions.
Stage 3b	Question cosine ≥ 0.90	Reject redundant question.

Table 2: Implemented verification, omission-recovery, and question-filtering rules.

437 claims and obtained a supported rate of 0.9680, unsupported claim rate of 0.0297, contradiction rate of 0.0252, and mean entailment of 0.9746. Sentence-first verification scored 146 claims and obtained 1.0000 supported rate, 0.0000 unsupported claim rate, 0.0000 contradiction rate, and 0.9972 mean entailment. This suggests that checking complete sentences before decomposition reduces unnecessary claim fragmentation in this development sample. The difference in the number of claims also means that the two strategies do not score an identical set of units.

7 Conclusion and Future Work

This project investigates a clinical documentation framework that combines interactive intake, evidence-grounded verification, and section-aware handling of clinical statements. We reproduced an adapted uMedSum baseline, implemented an Aegle-inspired multi-agent intake workflow, and developed an S/O verification pipeline with contradiction correction and validated omission recovery. On the 88-case test split, the proposed S/O pipeline improves atomic, critical, and recoverable fact recall while reducing unsupported and contradictory claims after verification. However, the current results do not yet demonstrate live clarification-based recovery or A/P evidence-chain verification.

The next steps are to complete the live patient clarification loop and validate responses before updating the note; conduct controlled RQ1 comparisons against deletion-only, guessing, and untargeted-question baselines; implement RQ2 evidence-chain verification with clinical-feature links and retrieved reference passages; and evaluate doctor-facing evidence flags. The final stage will generate and evaluate patient-facing summaries only after doctor sign-off. A matched evaluation setup is also required before making direct numerical comparisons between the uMedSum baseline and Verify-and-Ask.

Implementation Status

Repository

Code, datasets, and experiment outputs are organized in the project repository.

https://github.com/DoshikRahul/tokenizers

Pipeline stage	R-1	R-2	R-L	R-Lsum	BERT	SummaC	QuestEval	NLI	Words	Comp.
Initial 1-shot generation	0.4671	0.1852	0.2545	0.4278	0.8383	0.3298	0.3817	0.7969	353.2	0.298
Stage 2: RTB-TS removal	0.3365	0.1232	0.1820	0.2887	0.8264	0.3502	0.3503	0.9080	171.4	0.139
Stage 3: final uMedSum	0.3572	0.1294	0.1866	0.3001	0.8256	0.3452	0.3742	0.9018	201.4	0.162
Original paper, LLaMA-3-8B (A/P)	–	–	–	0.1700	0.8100	0.6300	0.4200	0.7700	∼140	∼0.090

Table 3: uMedSum baseline results on 140 full-SOAP encounters. R-1/2/L denote ROUGE scores; BERT denotes BERTScore F1; Comp. denotes compression ratio. The original-paper row is included only as contextual reference because its evaluation scope differs.

Metric	A: Stage 1	B: Stage 2	C: Stage 3a	Δ C–A
S/O ROUGE-1 F1	0.4533	0.4545	0.5044	+5.11 pts
S/O ROUGE-2 F1	0.2124	0.2132	0.2327	+2.03 pts
S/O ROUGE-Lsum F1	0.3980	0.3996	0.4377	+3.97 pts
S/O BERTScore F1	0.8767	0.8766	0.8783	+0.16 pts
Atomic fact recall	0.6334	0.6302	0.7342	+10.08 pts
Critical fact recall	0.6426	0.6380	0.7235	+8.09 pts
Recoverable fact recall	0.7101	0.7073	0.8199	+10.98 pts
Supported rate	0.9590	0.9862	0.9877	+2.87 pts
Unsupported claim rate	0.0334	0.0031	0.0051	−2.83 pts
Contradiction rate	0.0273	0.0000	0.0020	−2.53 pts
Mean entailment	0.9665	0.9891	0.9908	+2.43 pts
Number of claims	659	654	976	+317

Table 4: Verify-and-Ask results on the 88-case test split. A is the initial S/O note, B is the Stage 2 verified note, and C is the Stage 3a note after validated omission recovery.

Diagnostic	Dev (22)	Test (88)
Valid-claim retention proxy	100.00%	98.99%
Stage 2 claim retention	98.32%	98.52%
A/P unchanged rate	100.00%	100.00%
Genuine omissions	89	348
Validated additions	96	324
Gap-detection recall	30.34%	32.47%
Gap-detection precision	11.73%	13.89%
Fact recovery rate	39.33%	42.82%
Unsupported-addition rate	0.00%	0.00%
Duplicate-addition rate	6.25%	7.41%
Questions retained	25	88
Question-redundancy rate	28.57%	36.23%

Table 5: Stage-specific diagnostic metrics on the development and test splits.

Component	Status
uMedSum baseline reproduction	Completed
Aegle-inspired multi-agent intake	Completed
Stage 1 SOAP generation	Completed
Stage 2 S/O verification and correction	Completed
Stage 3a omission detection and recovery	Completed
Stage 3b question generation and filtering	Pending
RQ2 A/P evidence-chain verification	Pending
Doctor sign-off workflow	Pending
Patient-facing summary generation and evaluation	Pending
Matched baseline comparison	Pending

Table 6: Implementation status at the time of this report.